



MID SEMESTER EXAMINATION, AUTUMN-2024
School of Applied Sciences, KIIT-DU

B. Tech.
1st Semester (2024 AB)
Autumn 2024 (SAS)

Course: Science of Living System Code: LS10001 Full Marks: 20 Time: 90 minutes

Answer any **FOUR QUESTIONS** including **question No. 1** which is **compulsory**.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

		Marks	CO
Q.1			
a)	If possible, calculate the amount of cytosine in a single stranded DNA molecule with 1200 nucleotides where guanine is 25%.	1	2
b)	Recall one key difference between rough and smooth endoplasmic reticulum functions.	1	1
c)	In one line state the reason for chloroplasts being an exception to the second postulate of cell theory.	1	1
d)	Recall the type of RNA with acceptor and anticodon arms/loops.	1	2
e)	Define locus of a gene.	1	2
Q.2		Marks	CO
a)	In a double stranded molecule of DNA-RNA hybrid, there are 300 H-bonds between A and T. The number of U present is 200. If the percentage of G-C base pairs is 60%, calculate the length and number of pentose sugars present in the molecule.	3	2
b)	Explain the functions of nucleus and microtubules.	2	1
Q.3		Marks	CO
a)	Explain with diagrams, the applications of bacterial plasmids.	3	1
b)	Recall the types of chromosomes based on the position of the centromere.	2	2
Q.4		Marks	CO
a)	Describe the Central Dogma and the process of transcription which ends at the point of the processed mRNA transcript/mature mRNA. Use diagrams.	3	2
b)	Draw the diagram of a moneran cell.	2	1
Q.5		Marks	CO
a)	With the help of a diagram, explain packaging of DNA.	3	2
b)	Examine the statement: The genetic code is degenerate but non-ambiguous.	2	2
